

Class 02 — Teacher Cheat Sheet

Bong Study Hub — Foundation Batch 2026 · How Machines Learn From Rules → Data → Intelligence

Live-use operational reference, derived entirely from the approved Class 02 artifacts (Instructor Guide → Student Notes → Pen Tablet → Presentation → Interaction Pack → Homework → Assessment). **This is not a second Instructor Guide** — it's what you glance at mid-sentence. Full teaching language lives in the Master Instructor Guide.

1. Class at a Glance

Title	How Machines Learn
Subtitle	From Rules → Data → Intelligence
Duration	110 min canonical (90 / 120 variants — \$14–15)
Core question	"How can a machine make a decision when nobody explicitly wrote the rule for every situation?"
Core pipeline	DATA → LEARNING → MODEL → PREDICTION
Traditional pipeline	INPUT → HUMAN-WRITTEN RULES → OUTPUT
Primary outcome	Most students can explain <i>why</i> ML was needed for one real example, state the pipeline vs. rules, reason through cats-vs-dogs, and state that predictions aren't certainties and data quality matters.

CLASS NARRATIVE
A COMPUTER CAN FOLLOW RULES → BUT WHAT IF WE DON'T KNOW ALL THE RULES? → CAN WE GIVE IT EXAMPLES INSTEAD? →
DATA → LEARNING → MODEL → PREDICTION

3–5 ideas that matter most today:

- 1. The Rule Problem must be *felt*, not stated — protect the escalation.
 - 2. Machine Learning is named only *after* the rule problem lands.
 - 3. The Core Pipeline is built live, box by box — never shown finished first.
 - 4. A prediction is a best guess, not a guarantee.
 - 5. AI depends entirely on data — poor/narrow data → poor predictions.
- Class 01 callback:** PROBLEM → LOGIC → ALGORITHM → PROGRAM → RESULT and the learning staircase — today is a *preview from the top*, not a shortcut past the steps underneath.

DO NOT GO DEEPER TODAY: No code, no math, no implementation detail. See §11 for the full guardrail list.

2. Live Flow Table (110-min canonical)

Time	Section	Purpose	Ask / Do	Draw / Show	Watch For
0–8	Opening + AI perception	Activate existing AI experience	"Where have you noticed something 'smart' like this?" → state core question	None yet	"AI is just a robot" — bank it, don't correct yet
8–19	What is AI?	Working, non-philosophical notion	"Is a calculator AI? A thermostat?"	Light icon slide	Drifting into consciousness/philosophy debate
19–31	AI Around Us	Ground AI in 5 daily examples	"What does it receive? What does it produce?" per example	Drawing 1 (optional light list)	Claims about real products' internals
31–43	Traditional Programming	Re-anchor Input→Rules→Output	"What's your first rule?"	Drawing 2 — build live, empty → filled	Students sliding into code/ <code>if-else</code> syntax
43–59	Rule Problem (non-negotiable)	Feel the limitation directly	Escalating counter-examples → scaling Q → pivot Q	Drawing 3 — crowd the rule box	"Just need more rules" — let the crowding answer it
59–70	Introducing ML	Name ML only now	"What does 'learning a pattern' mean here?"	None yet (bare title)	"The computer thinks/understands" language
70–78	Core Pipeline (HERO, non-negotiable)	Install the pipeline live	4 questions, one per box	Drawing 4 — BUILD LIVE, never pre-drawn	"Patterns" as a 5th stage; Model/Prediction swap
78–90	Cats vs Dogs (HERO, non-negotiable)	Reason through the pipeline concretely	5-level question ladder (\$7)	None — point back to Drawing 4 if useful	Technical computer-vision vocabulary
90–95	Prediction ≠ Certainty (non-negotiable)	Best guess, not guarantee	"Was that a prediction or a guarantee?"	Point at Drawing 4's PREDICTION box	"If it's ML it should always be right"
95–99	AI Is Not Magic	Data quality → prediction quality	"What if it only ever saw English emails?"	Reuse Drawing 4, annotate DATA box	Reaching for "bias" as a buzzword with no reasoning
99–104	AI/ML/DL/GenAI Map	Orientation, not taxonomy	"Where does ChatGPT belong on this map?"	Drawing 5 — nested circles, ring by ring	Requests for neural-net/transformer detail
104–107	Return to Staircase	Preview from the top, not a shortcut	"Which chain have we built all class?"	Drawing 6 — quick callback, faster than Class 1's build	Framing today as contradicting Class 1's roadmap
107–110	Recap + Exit Check + Homework Bridge (non-negotiable)	Consolidate, verify, hand off	Recap + exit-check Qs (\$13)	None	Recited definitions without own-words explanation

3. Opening Script — First 5 Minutes

- Opening move:** "Quick show of hands — who used Instagram, YouTube, Netflix, or Google Maps in the last 24 hours?"
- First question:** "How many of you have actually thought about how it knows what to show you?" (*expect very few hands — that's the point*)
- State the core question, slowly:** "How can a machine make a decision when nobody explicitly wrote the rule for every situation?"
- Collect 2–4 quick answers**, without correcting any of them yet.
- Bridge:** "By the end of today, you'll be able to answer that yourself."

Expected responses: Instagram/YouTube/Netflix feed, Google Maps, ChatGPT, face unlock, autocorrect.

If silence: rephrase narrower within 5–7 seconds, or answer yourself first and toss it back ("I'll start us off — anyone else?").

Transition: straight into "What Is AI?" (\$2 row 2).

4. Core Explanation Cards

Card A — Traditional Programming

INPUT → HUMAN-WRITTEN RULES → OUTPUT

- **One-sentence:** A human thinks of the rule; the computer just follows it, exactly, every time.
- **Approved example:** Spam filter — "contains 'you have won' → spam."
- **Key question:** "What's your first rule?"
- **Watch for:** Sliding into code/ `if-else` syntax — keep it plain English.

Card B — Rule Problem

- **Students should notice:** The rule list keeps growing and still misses cases.
- **Role of variety / exceptions / change:** Real cases vary; spammers change wording; legitimate emails trip innocent-looking rules.
- **Approved example:** Legit — "*Your booking is urgent — confirm now!*" vs. Spam — "*fr33 amaz0n...*"
- **Pivot question:** "If humans cannot realistically write every rule, what could we give the computer instead?" (*wait 10–15 sec*)

Card C — Machine Learning

- **Conceptual explanation:** Give the system examples instead of rules, and let a learning process find useful patterns.
- **Framing phrase:** "Examples instead of every rule."
- **Wording care:** Say "notifies patterns," never "understands" or "thinks."
- **Transition into pipeline:** "Let's build out what 'learning from examples' looks like, step by step."

Card D — Core Pipeline (HERO CARD)

DATA → LEARNING → MODEL → PREDICTION

Stage	Means	You say	Students should say	Common confusion
DATA	Examples given to the system	"Past emails already labeled spam or not spam"	"The examples we give it"	Confusing data with rules
LEARNING	The process of finding useful patterns	"Looks across the examples, finds what separates spam from not-spam"	"It looks for patterns"	Thinking Learning is the model
MODEL	What the system builds from learning	"Something that's picked up the patterns"	"What it built after learning"	Confusing Model with Prediction
PREDICTION	The model's best guess on a new example	"Its best guess for a brand-new email"	"A guess, not a fact"	Treating prediction as guaranteed

"Patterns are what Learning finds — patterns are NOT a fifth stage."

MODEL = what the learning process builds. PREDICTION = what the model produces for something new.

Card E — Prediction ≠ Certainty

- **Core message:** A prediction is a best guess based on learned patterns — not a guarantee.
- **Cats/dogs connection:** Same as the ambiguous cat/dog guess — a reasonable guess that could still be wrong.
- **Watch for:** "If it's really ML, it should always be right."
- **Question:** "Was that a prediction, or a guarantee?"

Card F — Data Quality

- **Core message:** A model only knows what its examples showed it — poor or narrow data leads to poor predictions.
- **Pattern:** Narrow/one-sided examples → difficulty with unfamiliar cases (e.g., a spam filter that only ever saw English emails).
- **Do not require the word "bias."**
- **Question:** "What would happen if it only ever saw X?"

Card G — AI / ML / Deep Learning / GenAI

```
Artificial Intelligence
  ↓
Machine Learning
  ↓
Deep Learning
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Generative AI = a major modern application area, strongly associated with deep learning — shown touching the ML/Deep-Learning boundary, not as a strict fourth nested ring.

"This is a learning map, not a complete technical taxonomy."

DO NOT GO DEEPER: neural-network architecture, transformers, embeddings, tokens, LLM internals.

5. Pen Tablet Quick Cues

Drawing	When	Purpose	Build	Must Appear	Do NOT Pre-Draw	Transition
1 — AI Around Us	19–31 min	Ground AI in daily life	One line per named example, live	5-line flat list (spam, recs, maps, face unlock, GenAI)	The list itself	"None of these work by hand-writing a rule for every case."
2 — Traditional Programming	31–43 min	Human writes the rule	3-box skeleton, rules filled live by students	INPUT → HUMAN-WRITTEN RULES → OUTPUT + 3–4 rules	Any rules	"Let's throw some real emails at it."
3 — Rule Problem	43–59 min	Felt struggle	Crowd rules into/around Drawing 2's box, per counter-example	A visibly overcrowded box	A finished crowded box	Pivot question (spoken, not drawn)
4 — CORE PIPELINE	70–78 min	Install the pipeline	BUILD LIVE — DO NOT SHOW AS A FINISHED DRAWING BEFORE STUDENTS REASON TOWARD IT. One box per question, then stack Drawing 2's chain beneath for contrast	DATA → LEARNING → MODEL → PREDICTION + stacked comparison	Any part, at all, before the live build	Ground in spam example → "Let's reason through this with cats and dogs."
5 — AI/ML/DL/GenAI	99–104 min	Orientation map	3 nested circles ring by ring + GenAI star at the ML/DL boundary	3 rings + star, no 4th ring	The full map in advance	"Let's put today in context of Class 1's bigger picture."
6 — Staircase / Class 1 Callback	104–107 min	Preview from the top	Quick 8-step redraw, faster than Class 1's original build	8 labels: Programming → CS → Math → Data → ML → Deep Learning → LLMs → Agents	Month numbers / course names	Straight into Recap

Drawing 4 stays live and reachable from minute 70 through minute 107 — it's reused (pointed back to, never redrawn) for Prediction ≠ Certainty and AI Is Not Magic.

6. Activity Cues (Interaction Pack)

#	Activity	Time	Purpose	Prompt	Listen For	Move On When
1	AI Quick Reaction	5 min	Activate experience	"Which have you used? What makes something 'smart'?"	Ordinary-language answers ("it knows what I want")	3–4 answers collected
2	Build a Rule	6 min	Feel rule-based thinking	"What rule could identify spam?"	"Who decided this was a rule?" → "A human"	3–4 rules on board + authorship point made
3	Break the Rule	8 min	Create the need for a different approach	"Would our rules handle these correctly?"	Hesitation / patch attempts	2–3 challenges have landed
4	Rule Explosion	5 min	Make the growing-rule problem memorable	"If we keep doing this, what happens?"	"Too many rules / confusing / impossible"	"Rule explosion" label lands
5	Pivot Question	4 min	Let students arrive at "data" themselves	"What could we give the computer instead?" — STOP	"Examples / data / past cases"	Only after a student-sourced answer (wait 5–10 sec min.)
6	Predict the Pipeline	8 min	Construct the pipeline live	4 sequential questions (Card D)	Box-appropriate answers	All 4 boxes built + grounded in spam example
7	Cats vs Dogs	12 min	See §7 — HERO activity	—	—	—
8	Prediction or Certainty?	5 min	Correct "AI = always right"	"Prediction, or guarantee?"	"Prediction"	Best-guess-not-guarantee stated
9	Data Quality Thought Experiment	4 min	Poor data → poor predictions	"Small white cats / large black dogs → large white cat?"	"Confused / wrong pattern"	"Does data matter?" → "Yes" lands
10	AI/ML/DL/GenAI Check	4 min	Orientation check	"Which is broader — AI or ML? Where's GenAI?"	Broad-to-narrow reasoning	Map completed + "not a taxonomy" reminder given
11	Exit Check	5 min	Formative check	4 questions (§13)	Own-words explanation vs. recited definition	Most students have attempted

7. Hero Activity — Cats vs. Dogs

Purpose: Reason through the Machine Learning idea via a visual classification example — no code, no math.

Setup: Many labeled cat examples + many labeled dog examples, deliberately varied (breed, color, pose, background). Presentation Slide 17.

Ask (question ladder):

1. Observation — "What do you notice?"
2. Comparison — "What seems similar across the cat examples? Across the dog examples?"
3. Generalization — "If the next cat looks very different, could the system still recognize it?"
4. Ambiguity — "What if the new image is difficult to classify?"
5. Limitation — "Could the system make a mistake?"

Wait: 5–10 seconds per question — longer than usual.

Follow-up: "Now imagine a brand-new image — what would the system do?" → "Could it still get it wrong?" → **Yes.**

What good reasoning sounds like: Names a visual pattern (ear shape, size, face shape) **and** connects it to "if it saw enough examples, it could notice this too" **and** acknowledges the guess could still be wrong.

Common wrong answers: A hand-written-rule answer ("if pointy ears → cat"); technical computer-vision jargon; assuming guaranteed correctness.

How to handle a wrong answer: Don't grade "correct features" — accept any reasonable pattern. Redirect a rule-based answer gently toward "pattern from examples" framing, not "wrong, try again."

When to move on: Once "prediction, could be wrong" is reached — this is the direct bridge into Prediction ≠ Certainty.

Pen-tablet / slide connection: No new drawing. If useful, point back to Drawing 4 and trace the same four boxes with the cat/dog framing.

8. Misconception Radar

Misconception	Listen For	Response / Reframe
"AI is just a robot / ChatGPT."	"AI means robots," or reducing AI to one product	"AI mostly lives inside ordinary apps you already use — no body required."
"Surely we could just write enough rules to cover it."	"We just need more rules"	Let the crowding on Drawing 3 answer this visually — "and there's always another case waiting."
"Rules are difficult only because writing them takes time."	Treating the Rule Problem as an effort problem	"It's not about how long it takes — it's that the situations and exceptions keep growing faster than any list can."
"Machine Learning means no programming is needed."	"So we don't need programmers anymore?"	"Someone still has to build, prepare data for, and deploy these systems — ML is an extra tool, not a replacement."
"Patterns are a separate, fifth pipeline stage."	Student adds "patterns" as its own box	"Patterns are what Learning <i>finds</i> — the pipeline stays four stages."
"Model and Prediction are the same thing."	Swapping the two in an explanation	"Model = what the learning process builds. Prediction = what the model produces for something new."
"A prediction means certainty."	"If it predicted it, it must be right"	"A prediction is a best guess based on learned patterns — not a guarantee."
"If ML makes a mistake, it wasn't really using ML."	"That can't be real Machine Learning, it got it wrong"	"A wrong guess doesn't mean the system is broken — reasonable guesses can still be wrong."
"AI thinks/understands like a human."	"The AI knows/understands X"	"Think of it as noticing patterns, not understanding the way you do."
"More data automatically means better AI."	"Just give it more data and it'll be better"	"More <i>good, relevant</i> data usually helps — messy or irrelevant data can teach the wrong patterns faster."

Address these opportunistically as they surface — never read this table aloud as a block, and never let a correction feel like a callout of the student personally.

9. Quiet-Class Protocol

ASK

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WAIT 3-5 SEC (5-10 sec for the Pivot Question and Cats-vs-Dogs)

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NO RESPONSE?

↓

MAKE THE QUESTION EASIER / NARROWER

↓

TRY CHAT, HAND-RAISE, OR A QUICK PAIR CHECK

↓

TAKE ONE RESPONSE

↓

BUILD FROM IT — DO NOT ANSWER IT YOURSELF

Example narrowing: instead of "What pattern should the system learn?" ask "Do you think it should look at the words, the sender, or both?"

10. Advanced-Student Parking Lot

If a student raises any of the following, **do not answer technically** — acknowledge and park it:

- Neural-network math
- Gradient descent / optimization
- Transformers
- Embeddings / tokens
- RAG
- Agents (as a technical mechanism)
- Formal ML metrics (accuracy, precision, recall)
- Model-training implementation
- Deep prompt engineering

Standard response:

"Good question. We're going to build toward that later. Today we're staying with the mental model."

Then return directly to DATA → LEARNING → MODEL → PREDICTION . Don't let one advanced student pull the whole class into a later topic.

11. DO NOT GO DEEPER TODAY

- No code, no pseudocode, no programming syntax
- No mathematics, no formulas
- No implementation details
- No proprietary product internals (don't claim to know how a real company's system actually works)
- No neural-network architecture explanation
- No formal ML terminology beyond class level (no probability, confidence scores, accuracy/precision/recall, loss functions, gradient descent, transformers, embeddings, RAG)
- "Agents" may appear **only** as a future staircase-step label — never explained

Glance here the moment discussion starts drifting technical.

12. Transition Cues

From	To	Say
Rules (built)	Rule Problem	"Let's throw some real emails at it."
Rule Problem	Pivot	"So if writing every rule becomes impractical, what else could we give the computer?"
Pivot (answered)	Machine Learning	"That instinct has a name."
Machine Learning	Core Pipeline	"Let's build out what 'learning from examples' looks like, step by step."
Core Pipeline	Cats vs Dogs	"Let's reason through this with something concrete: cats and dogs."
Cats vs Dogs	Prediction ≠ Certainty	"Could it still get it wrong?" → (yes)
Prediction ≠ Certainty	AI Is Not Magic	"This connects to something bigger: AI isn't magic."
AI Is Not Magic	AI/ML/DL/GenAI Map	"Let's zoom out and see how all the AI terms you've heard fit together."
AI Map	Staircase	"Let's put today in context of the bigger picture from Class 1."
Staircase	Recap	(straight in — no new drawing or slide)

13. Final 5 Minutes

- ☐ Return to the core question — ask students to answer it now, in their own words.
- ☐ Restate the core pipeline: DATA → LEARNING → MODEL → PREDICTION .
- ☐ Ask the exit-check questions: 1. Why might hand-written rules become difficult? 2. What is the alternative? 3. What is the four-stage pipeline? 4. Why isn't prediction certainty? 5. Why does the data matter?
- ☐ Connect back to Class 1 — the staircase, "preview from the top."
- ☐ Preview what comes next — Month 1/2 foundations build the steps underneath the staircase.
- ☐ Point to the **Homework** package (06_Homework/) — 10 questions, 30–45 min, no coding.
- ☐ If assigned, point to the **Assessment** (07_Assessment/) — 7 questions, 25–35 min, fresh scenarios.

Accept answers in the student's own words — this is a formative close, not a lecture.

14. Emergency Time-Cut Version (~90 Minutes)

Always protect: Rule Problem · Pivot · Core Pipeline · Cats-vs-Dogs reasoning · Prediction ≠ Certainty · AI/ML/DL/GenAI map · Recap.

Compress:

- Opening + AI perception → 2–3 examples collected instead of open-ended.
- What Is AI? → skip the calculator/thermostat debate, state the working idea directly.
- AI Around Us → 3 examples (spam, recommendations, face unlock) instead of 5.
- Traditional Programming → cap rule collection at 3 rules.
- Rule Problem → 3 escalating counter-examples instead of 4–5 — the felt struggle must still land.
- Introducing ML → one clean statement + one check-in question.
- Cats vs Dogs → drop the extension round only; keep the full core sequence.
- AI Is Not Magic → one example only (language), not a second.

Last-resort further cut (only if still over time after the Rule Problem): deliver the AI/ML/DL/GenAI map as a 2-minute **spoken** version instead of live-drawn — never cut the Core Pipeline or the Recap.

15. Extended Version (~120 Minutes)

Every added minute goes to **interaction and reasoning**, never new technical content:

- Opening → one more student describes their example in more detail.
 - AI Around Us → add a 6th example students suggest themselves.
 - Rule Problem → let students try to patch their own rules for 2–3 more rounds before the reveal.
 - Cats vs Dogs → add a stronger-student extension round (a harder ambiguous example, student-proposed).
 - AI/ML/DL/GenAI Map → 1–2 extra "where does X belong?" questions (e.g., self-driving cars, voice assistants).
 - Staircase → only if time genuinely allows, redraw the Traditional-vs-ML comparison next to the staircase instead of pointing back to Drawing 4.
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16. Pre-Class 2-Minute Checklist

- ☐ Presentation ready (03_Presentation/Class_02_How_Machines_Learn.pptx)
- ☐ Pen tablet / canvas ready — 5 pages set up (see \$5)
- ☐ Core Pipeline page (Drawing 4) reachable and empty — not pre-drawn
- ☐ Cats-vs-Dogs example images/descriptions ready
- ☐ Interaction Pack open for quick reference
- ☐ Homework package ready to share (06_Homework/)
- ☐ Assessment ready if applicable (07_Assessment/)

- ☐ No advanced material planned or prepared
 - ☐ Remember: **ask before explaining**
-

17. Post-Class Capture

Not the full Reflection artifact — that belongs in 10_Reflection/. This is just a quick note to yourself before you forget.

TODAY I NOTICED:

Most common misconception: _____

Question students struggled with: _____

Question that produced strong reasoning: _____

Timing that ran long: _____

What to revisit next class: _____